

Situation

Nordwerk Industrievertretung (illustrative mid-size industrial distributor) runs a 12-person inside-sales desk that hand-builds quotes from ~2,000 unstructured RFQ / order emails a week — read each one, match every line to a SKU, check stock and price, type back a quote.

The problem, in numbers

17.3k h

inside-sales hours / year
on manual quoting

€780k

fully-loaded labour cost
/ year (estimate)

~3%

assumed hand-keying
error rate on lines

next-day

typical turnaround
under load

Solution

A from-scratch LLM tool-use loop that, given one email, decides its own steps and drafts the quote. Off-catalogue items are flagged for a human, never guessed. Tested Python underneath; an optional n8n front-door logs every run — the desk reviews and sends.

parse_email

lookup_sku ×5

check_stock ×5

draft_quote

Measured on a real 5-line email: 12 tool calls across 13 steps → one drafted quote.

Impact / ROI

Conservative case: a rep still reviews and approves every draft in ~2 min instead of keying for ~10 min — an 80% cut, before any straight-through automation.

80%

time cut per quote
(10 min → 2 min)

13.9k h

inside-sales hours saved
/ year

€624k

labour saved / year
(estimate)

<2 mo

payback on ~€100k
build (estimate)

Recommendation

Pilot the RFQ agent on one product division as tested full-code tool logic behind an n8n operator front-door — the hybrid the decision guide recommends. Measure live model cost and latency on real emails before scaling to the full desk.

How the numbers were derived

Assumptions (estimates)

RFQ / order emails per week	2,000
Manual handling per email	10 min
Review-and-send per email (automated draft)	2 min
Fully-loaded inside-sales labour cost	€45 / hour
Working weeks per year	52
One-off build + integration (estimate)	€100,000

The arithmetic

Manual hours/year = $2,000 \times 10 \text{ min} \div 60 \times 52 \approx 17,333 \text{ h}$

Manual cost/year = $17,333 \text{ h} \times €45 \approx €780,000$

Time cut per quote = $(10 - 2) \div 10 = 80\%$

Hours saved/year = $17,333 \times 80\% \approx 13,867 \text{ h}$

Cost saved/year = $13,867 \text{ h} \times €45 \approx €624,000$

Payback = $€100,000 \div (€624,000 \div 12) \approx 1.9 \text{ months}$

Measured behaviour (not estimated)

From benchmarks/results/metrics.json, 25 runs of the flagship agent on a real 5-line email:

- 12 tool calls across 13 steps, converging to one drafted quote
- tool sequence: parse_email → lookup_sku ×5 → check_stock ×5 → draft_quote

The number of tool calls tracks the number of line items — that is what makes it agentic, not a fixed script.

Read more

- docs/BUSINESS_CASE.md — the full one-page business case
- docs/DECISION_GUIDE.md — low-code vs full-code, per use case
- benchmarks/results/scorecard.md — the measured 9-dimension scorecard